- The cloud is a dynamic environment, and the topology keeps changing. Service discovery and load balancing let you dynamically establish interactions between services, managed either on the client side (for example, using Spring Cloud Netflix Eureka) or on the server side (for example, using Kubernetes).
- Kubernetes provides a native service-discovery and load-balancing feature that you can use through the Service objects.
- Each Service name can be used as a DNS name. Kubernetes will resolve the name to the Service IP address and, ultimately, forward the request to one of the instances available.
- You can deploy Spring Boot applications to a Kubernetes cluster by defining two YAML manifests: one for the Deployment object and one for the Service object.
- The kubectl client lets you create objects from a file with the command `kubectl apply -f <your-file.yml>`.
- Cloud native applications should be disposable (fast startup and graceful shutdown) and stateless (rely on data services for storing the state).
- Graceful shutdown is supported both by Spring Boot and Kubernetes and is an essential aspect of scalable applications.
- Kubernetes uses ReplicaSet controllers to replicate your application Pods and keep them running.
- Tilt is a tool that automates your local development workflow with Kubernetes: you work on the application while Tilt takes care of building the image, deploying it to your local Kubernetes cluster, and keeping it up-to-date whenever you change something in the code.
- You can start Tilt for your project with `tilt up`.